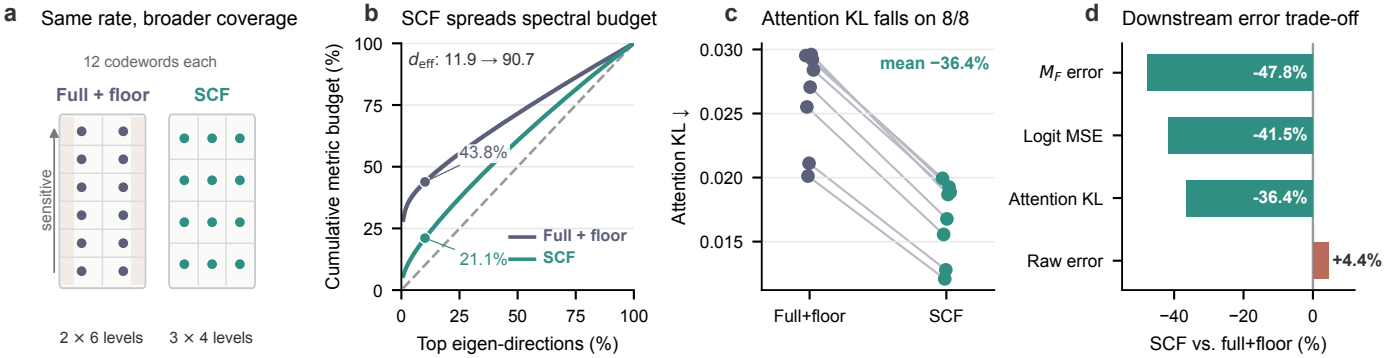
a
Same rate, broader coverage
12 codewords each
Full + floor
SCF
sensitive
2 × 6 levels
3 × 4 levels
b
SCF spreads spectral budget
d_{eff}: 11.9 → 90.7
43.8%
21.1%
Full + floor
SCF
Cumulative metric budget (%)
Top eigen-directions (%)
c
Attention KL falls on 8/8
mean −36.4%
Attention KL ↓
Full+floor
SCF
d
Downstream error trade-off
M_F error
Logit MSE
Attention KL
Raw error
-47.8%
-41.5%
-36.4%
+4.4%
SCF vs. full+floor (%)